

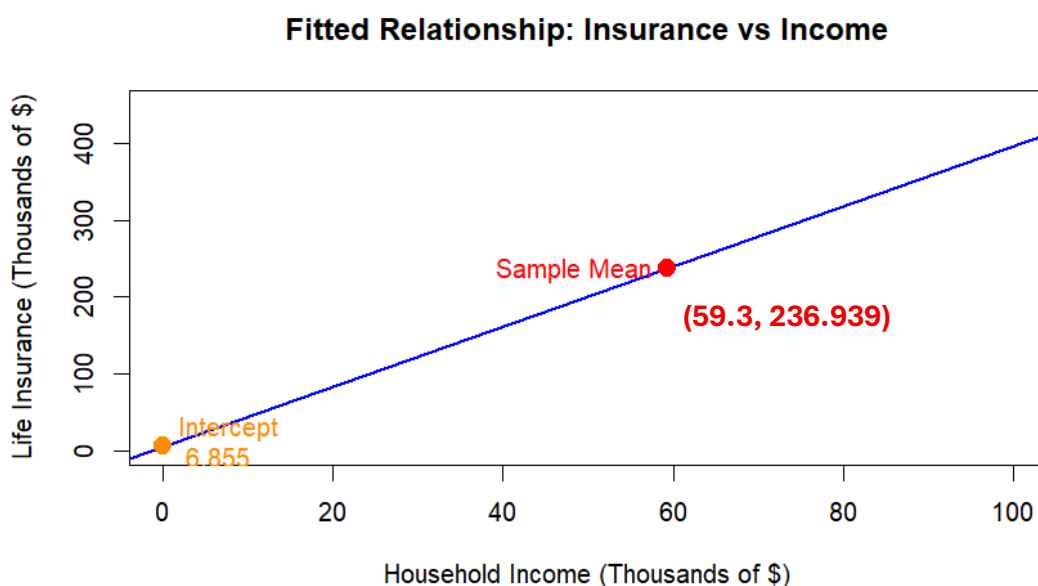
- 3.18** A life insurance company examines the relationship between the amount of life insurance held by a household and household income. Let *INCOME* be household income (thousands of dollars) and *INSURANCE* the amount of life insurance held (thousands of dollars). Using a random sample of $N = 20$ households, the least squares estimated relationship is

$$\widehat{INSURANCE} = 6.855 + 3.880INCOME$$

(se) (7.383) (0.112)

- a. Draw a sketch of the fitted relationship identifying the estimated slope and intercept. The sample mean of *INCOME* = 59.3. What is the sample mean of the amount of insurance held? Locate the point of the means in your sketch.
- b. How much do we estimate that the average amount of insurance held changes with each additional \$1000 of household income? Provide both a point estimate and a 95% interval estimate. Explain the interval estimate to a group of stockholders in the insurance company.
- c. Construct a 99% interval estimate of the expected amount of insurance held by a household with \$100,000 income. The estimated covariance between the intercept and slope coefficient is -0.746 .
- d. One member of the management board claims that for every \$1000 increase in income the average amount of life insurance held will increase by \$5000. Let the algebraic model be $INSURANCE = \beta_1 + \beta_2 INCOME + e$. Test the hypothesis that the statement is true against the alternative that it is not true. State the conjecture in terms of a null and alternative hypothesis about the model parameters. Use the 5% level of significance. Do the data support the claim or not? Clearly, indicate the test statistic used and the rejection region.
- e. Test the hypothesis that as income increases the amount of life insurance held increases by the same amount. That is, test the null hypothesis that the slope is one. Use as the alternative that the slope is larger than one. State the null and alternative hypotheses in terms of the model parameters. Carry out the test at the 1% level of significance. Clearly indicate the test statistic used, and the rejection region. What is your conclusion?

(a)



Calculate the mean: The OLS regression line must pass through the sample mean point $(\bar{X}, \bar{Y})$. Substituting $INC\bar{O}ME = 59.3$ into the equation:

$$\overline{INSURANCE} = 6.855 + 3.880 \times 59.3 = 6.855 + 230.084 = 236.939$$

The sample mean of insurance amount is \$236.939 (thousand dollars).

(b)

As the slope of the regression line is 3.880, the average amount of insurance held will increase $3.880 \times 1,000 = \$3,880$ with each additional 1,000 of household income.

- **95% Confidence Interval:** From the t-table, the critical value for a two-tailed 5% significance level with 18 degrees of freedom is $t_{0.025,18} = 2.101$. The interval formula: $\hat{\beta}_2 \pm t \times se(\hat{\beta}_2)$

$$3.880 \pm 2.101 \times 0.112 = 3.880 \pm 0.2353$$

The 95% confidence interval is [3.6447, 4.1153] (thousand dollars), which converts to [3,644.7, 4,115.3] dollars.

- It means that we have 95% confident that the average amount of insurance held will increase at an amount between 3,644.7 and 4,115.3 with each additional 1,000 of household income.

```
> lower_bound
[1] 3.644697
> upper_bound
[1] 4.115303
```

(c)

$$INSURANCE = 6.855 + 3.880 \times INCOME = 6.855 + 3.880 \times 100 = 394.855$$

```
> #calculate point estimator
> insurance_hat = intercept + slope*100
> print(insurance_hat)
[1] 394.855
```

$$\sqrt{\text{var}(b1 + 100b2)} = \sqrt{\text{var}(b1) + 100^2\text{var}(b2) + 2 \times 100 \times \text{cov}(b1, b2)}$$

$$= \sqrt{7.383^2 + 10000 \times 0.112^2 + 200 \times (-0.746)} = \sqrt{30.748689} = 5.5452$$

```
> #calculate SE
> variance <- b_1^2 + 100^2 * b_2^2 + 2 * 100 * covariance
> standard_error <- sqrt(variance)
> print(standard_error)
[1] 5.54515
```

the critical value for a two-tailed 5% significance level with 18 degrees of freedom is

$$t_{0.005, 18} = 2.878$$

99% interval estimate of the expected amount of insurance held by a household with a \$100,000 income: $394.855 \pm 2.878 \times 5.5452 = [378.8959, 410.8141]$

```
> #calculate C.I.
> upper_bound <- insurance_hat + 2.878 * standard_error
> lower_bound <- insurance_hat - 2.878 *
+ print(upper_bound)
[1] 410.8139
> print(lower_bound)
[1] -787.4675
```

In repeated sampling, we estimate with a 99% confidence level that the expected insurance holdings of households with an income of \$100,000 are between \$378.8959 and \$410.8141.

(d)

① For every \$1000 increase in income, the average amount of life insurance held increases by \$5000.

Null Hypothesis $H_0: \beta_2 = 5$

Alternative Hypothesis $H_1: \beta_2 \neq 5$

② significance level of 5%

③ We will test the hypothesis using a t-test.

The test statistic is calculated as: $t = \frac{b_2 - 5}{se(b_2)} = \frac{3.88 - 5}{0.112} = -10$

④ Rejection Region: $\{ t \leq -2.101 \text{ or } t \geq 2.101 \}$

the critical value for a two-tailed 5% significance level with 18 degrees of freedom is $t_{0.025,18} = 2.101$.

⑤ -10 is less than -2.101, t falls in rejection region. We **reject** the null hypothesis and conclude that the data do not support the claim which for every \$1000 increase in income the average amount of life insurance held will increase by \$5000.

(e)

①

Null Hypothesis $H_0: \beta_2 = 1$ (The amount of life insurance held increases by the same amount as income increases)

Alternative Hypothesis $H_1: \beta_2 > 1$ (The amount of life insurance held increases doesn't held by the same amount as income increases)

② significance level of 1%

③ We will test the hypothesis using a t-test.

The test statistic is calculated as: $t = \frac{b_2 - 1}{se(b_2)} = \frac{3.88 - 1}{0.112} = 25.7143$

④ Rejection Region: $\{ t \geq 2.552 \}$

the critical value for a right-tail 1% significance level with 18 degrees of freedom is $t_{0.01,18} = 2.552$.

⑤ Because t falls in rejection region , we **reject** the null hypothesis and conclude that the data do not support the claim. It indicates that the **slope is larger than 1** .

3.23 The data file *collegetown* contains data on 500 single-family houses sold in Baton Rouge, Louisiana, during 2009–2013. The data include sale price in \$1000 units, *PRICE*, and total interior area in hundreds of square feet, *SQFT*.

- Using the quadratic regression model, $PRICE = \alpha_1 + \alpha_2 SQFT^2 + e$, test the hypothesis that the marginal effect on expected house price of increasing the size of a 2000 square foot house by 100 square feet is less than or equal to \$13000 against the alternative that the marginal effect will be greater than \$13000. Use the 5% level of significance. Clearly state the test statistic used, the rejection region, and the test *p*-value. What do you conclude?
- Using the quadratic regression model in part (a), test the hypothesis that the marginal effect on expected house price of increasing the size of a 4000 square foot house by 100 square feet is less than or equal to \$13000 against the alternative that the marginal effect will be greater than \$13000.

Use the 5% level of significance. Clearly state the test statistic used, the rejection region, and the test *p*-value. What do you conclude?

- Using the quadratic regression model in part (a), estimate the expected price $E(PRICE|SQFT) = \alpha_1 + \alpha_2 SQFT^2$ for a house of 2000 square feet. Construct a 95% interval estimate of the expected price. Describe your interval estimate to a general audience.
- Locate houses in the sample with 2000 square feet of living area. Calculate the sample mean (average) of their selling prices. Is the sample average of the selling price for houses with $SQFT = 20$ compatible with the result in part (c)? Explain.

(a)

The estimated model is $\widehat{PRICE} = 93.5659 + 0.1845 SQFT^2$

Call:

```
lm(formula = price ~ I(sqft^2), data = collegetown)
```

Residuals:

Min	1Q	Median	3Q	Max
-383.67	-48.39	-7.50	38.75	469.70

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	93.565854	6.072226	15.41	<2e-16 ***
I(sqft^2)	0.184519	0.005256	35.11	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 92.08 on 498 degrees of freedom

Multiple R-squared: 0.7122, Adjusted R-squared: 0.7117

F-statistic: 1233 on 1 and 498 DF, p-value: < 2.2e-16

the marginal effect is $\frac{d PRICE}{d SQFT} = 2\hat{\alpha}_2 SQFT = 2\hat{\alpha}_2(20) = 7.3808$

Standard error is $\sqrt{\widehat{var}(40\hat{\alpha}_2)} = 40(se(\alpha_2)) = 0.2102$

① Null Hypothesis $H_0: 2\alpha_2(20) \leq 13$

Alternative Hypothesis $H_1: 2\alpha_2(20) > 13$

② significance level of 5%

③ This is a right-tail test. The test statistic is $t = \frac{\hat{\alpha}_2 - 13}{se(\hat{\alpha}_2)} = \frac{7.3808 - 13}{0.2102} = -26.7288$

④ Rejection Region: { $t \geq 1.6479$ }

the critical value for a right-tail 5% significance level with (500-2) degrees of freedom is $t_{0.95, 498} = 1.6479$.

⑤ Because t does not fall in rejection region, we **cannot reject** the null hypothesis.

We cannot conclude that the marginal effect is greater than \$13,000.

p-value is 1.

```
>
> # margin effect and std
> aphla2 = coef(model)[[2]]
> margine_effect = 2*aphla2*20
> sd = 40*(0.005256)
> # test
> t = (margine_effect-13)/sd
> df = dim(collegetown)[1]-2
> p = 1 - pt(q = t, df = df, lower.tail = T)
> print(paste("t-value =", t))
[1] "t-value = -26.7277391681723"
> print(paste("df =", df))
[1] "df = 498"
> print(paste("p-value =", p))
[1] "p-value = 1"
```

(b)

The estimated model is $\widehat{PRICE} = 93.5659 + 0.1845 SQFT^2$

the marginal effect is $\frac{d PRICE}{d SQFT} = 2\hat{\alpha}_2 SQFT = 2\hat{\alpha}_2(40) = 14.762$

Standard error is $\sqrt{\widehat{var}(80\hat{\alpha}_2)} = 80(se(\alpha_2)) = 0.4205$

① Null Hypothesis $H_0: 2\alpha_2(20) \leq 13$

Alternative Hypothesis $H_1: 2\alpha_2(20) > 13$

② significance level of 5%

③ This is a right-tail test. The test statistic is $t = \frac{\hat{\alpha}_2 - 13}{se(\hat{\alpha}_2)} = \frac{14.762 - 13}{0.4205} = 4.189$

④ Rejection Region: $\{ t \geq 1.6479 \}$

the critical value for a right-tail 5% significance level with (500-2) degrees of freedom is $t_{0.95, 498} = 1.6479$.

⑤ Because t falls in rejection region, we **reject** the null hypothesis.

The marginal effect is larger than \$13000.

p-value is 0.00001654.

```
I(sqft^2)
14.76152
> se
[1] 0.4204641
> t_statistic_2
I(sqft^2)
4.189467
> critical_2
[1] 1.647919
> pvalue_2
I(sqft^2)
1.65395e-05
```

(c)

$$E(PRICE|SQFT) = \alpha_1 + \alpha_2 \times SQFT^2$$

100(1- α)% interval estimate for $(c_1\beta_1 + c_2\beta_2)$

$$is(c_1b_1 + c_2b_2) \pm t_c se(c_1b_1 + c_2b_2)$$

The expected price is

$$\hat{(E)}(PRICE|SQFT = 20) = 93.5659 + 0.1845 * 20^2 = 167.3735$$

The standard error is

$$\sqrt{\text{var}(\hat{\alpha}_1) + 400 * (\hat{\alpha}_2)} = \sqrt{\text{var}(\hat{\alpha}_1) + 400^2 * (\hat{\alpha}_2) + 2(400)\text{cov}((\hat{\alpha}_1), (\hat{\alpha}_2))}$$

$$= 4.746378$$

95% Confidence Interval: From the t-table, the critical value for a two-tailed 5% significance level with 498 degrees of freedom is $t_{0.975, 498} = 1.9648$.

$$167.3735 \pm (1.9648) 4.746378 = [158.0481, 176.6988]$$

The 95% confidence interval is **[158.0481, 176.6988]**

```
> interval
(Intercept) (Intercept)
  158.0481    176.6988
```

In repeated sampling, we estimate with 95% confidence that the average price of a 2000 square foot house is between \$158,048.1 and \$176,698.8.

(d)

In the sample there are 3 houses with 2000 square feet (SQFT = 20). They sold for **\$138,000**, **\$169,000** and **\$183,000** respectively, and the sample average is 163.3, which is inside the interval estimate [158.0481, 176.6988].

```
num [1:3] 138 169 183
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  138.0   153.5   169.0   163.3   176.0   183.0
```

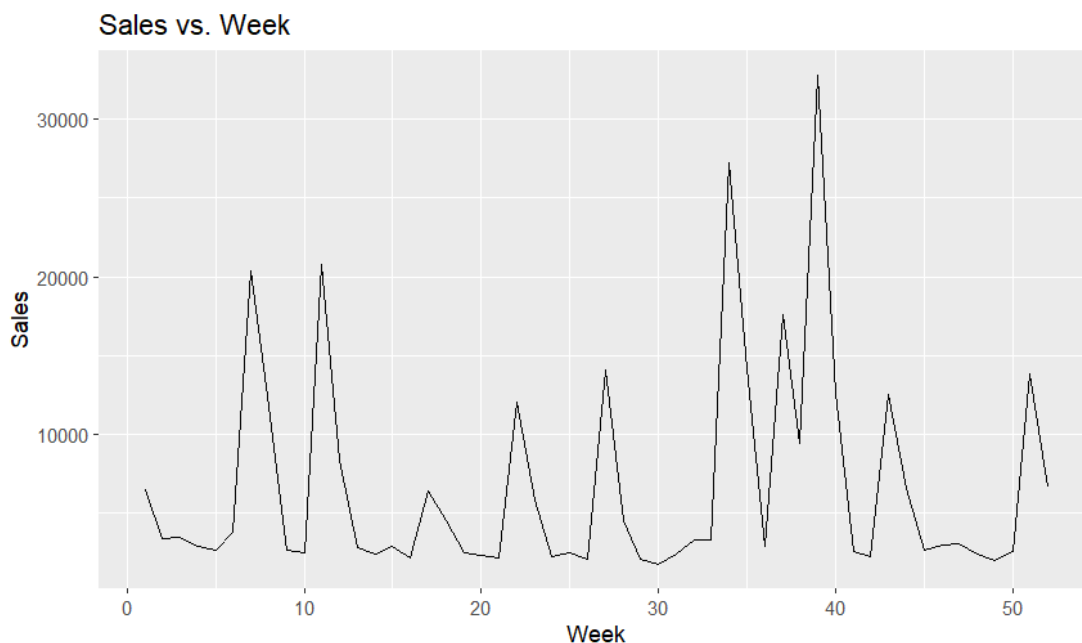
This value closely approximates the expected price from (c) and should lie within its 95% confidence interval. This is because the OLS regression line is designed to capture the conditional expectation $E(Y|X)$.

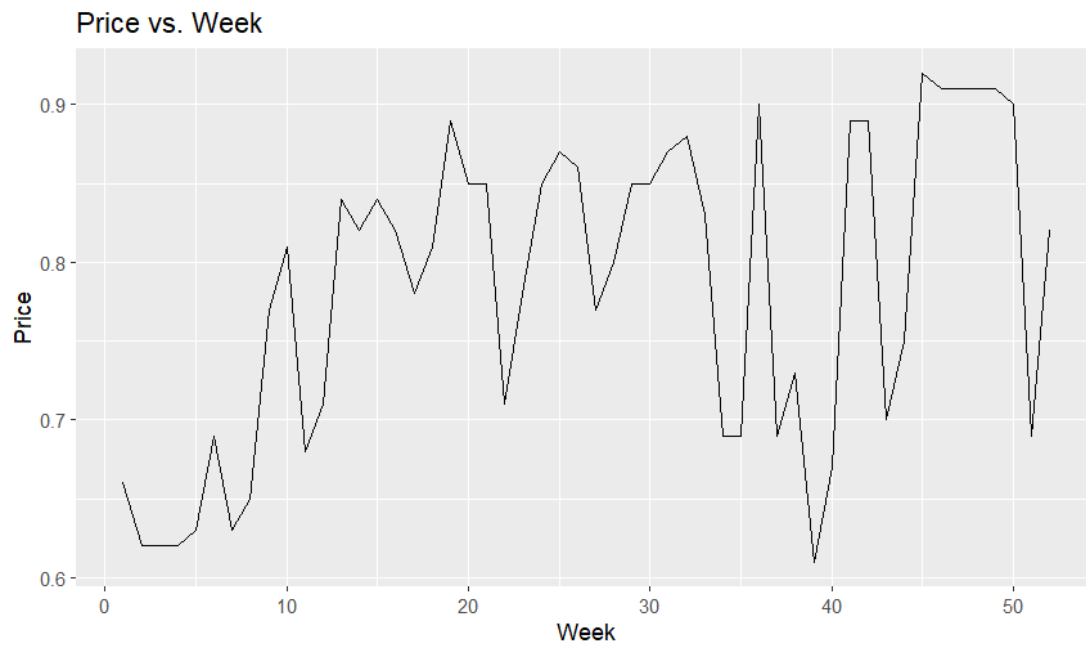
3.31 Data on weekly sales of a major brand of canned tuna by a supermarket chain in a large midwestern U.S. city during a mid-1990s calendar year are contained in the data file *tuna*. There are 52 observations for each of the variables. The variable *SAL1* = unit sales of brand no. 1 canned tuna, and *APR1* = price per can of brand no. 1 tuna (in dollars).

- Calculate the summary statistics for *SAL1* and *APR1*. What are the sample means, minimum and maximum values, and their standard deviations. Plot each of these variables versus *WEEK*. How much variation in sales and price is there from week to week?
- Plot the variable *SAL1* (y-axis) against *APR1* (x-axis). Is there a positive or inverse relationship? Is that what you expected, or not? Why?
- Create the variable $PRICE1 = 100APR1$. Estimate the linear regression $SAL1 = \beta_1 + \beta_2 PRICE1 + e$. What is the point estimate for the effect of a one cent increase in the price of brand no. 1 on the sales of brand no. 1? What is a 95% interval estimate for the effect of a one cent increase in the price of brand no. 1 on the sales of brand no. 1?
- Construct a 90% interval estimate for the expected number of cans sold in a week when the price per can is 70 cents.
- Construct a 95% interval estimate of the elasticity of sales of brand no. 1 with respect to the price of brand no. 1 "at the means." Treat the sample means as constants and not random variables. Do you find the sales are fairly elastic, or fairly inelastic, with respect to price? Does this make economic sense? Why?
- Test the hypothesis that elasticity of sales of brand no. 1 with respect to the price of brand no. 1 from part (e) is minus three against the alternative that the elasticity is not minus three. Use the 10% level of significance. Clearly, state the null and alternative hypotheses in terms of the model parameters, give the rejection region, and the *p*-value for the test. What is your conclusion?

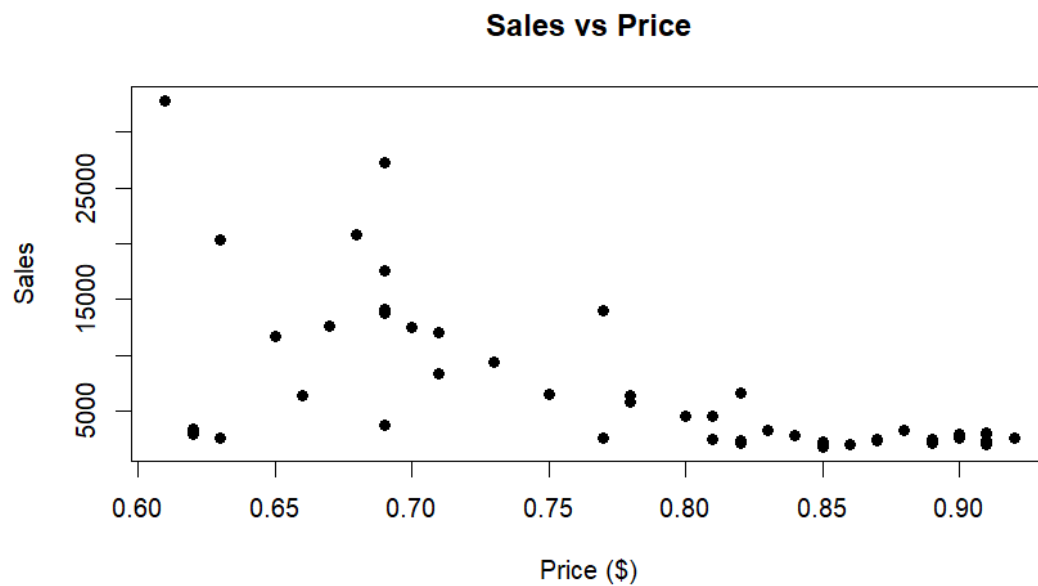
(a)

Variable	Sample_ Mean	Minimum_ Value	Maximum_ Value	Standard_ Deviation
SAL1	6718.71	1762.00	32820.00	6915.08
APR1	0.78	0.61	0.92	0.10





(b)



This scatter plot shows an inverse relationship between sales and price.

When the price per canned tuna rises, the sales goes down.

This is consistent with the law of demand: when the price of a good rises,

demand becomes lower ,consumers' willingness to buy and sales decrease.

(c)

The effect of a 1 cent increase in the price means the margin effect of *PRICE1* , that is

$$\frac{\partial SAL1}{\partial PRICE1} = \beta_2.$$

The point estimator for β_2 is $b_2 = -434.4473$.

$$SE(b_2) = \frac{\hat{\sigma}^2}{\sum(x_i - \bar{x})^2} = 78.5848$$

- **95% Confidence Interval:** From the t-table, the critical value for a two-tailed 5% significance level with 50 degrees of freedom is $t_{0.025,50} = 2.0086$.

$$-434.4473 \pm (2.0086) 78.5848 = [-592.2897, -276.6049]$$

The 95% confidence interval is $[-592.2897, -276.6049]$

```
> lower_bound  
[1] -592.2897  
> upper_bound  
[1] -276.6049
```

- It means that we have 95% confident that when one cent increases in the price of brand no.1 , the average sale of tuna cans in a week will be decreased between 592.2897 and 276.6049

(d)

$$SAL1 = 40714.22 - 434.45 \times PRICE1$$

```

Residuals:
    Min       1Q   Median       3Q      Max
-10869.5  -1588.3   -499.3   1626.2  18607.1

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  40714.22    6196.42    6.571 2.82e-08 ***
price1       -434.45     78.58   -5.528 1.17e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5502 on 50 degrees of freedom
Multiple R-squared:  0.3794,    Adjusted R-squared:  0.367
F-statistic: 30.56 on 1 and 50 DF,  p-value: 1.172e-06

```

Given $X_0 = 70$, we can obtain $\hat{Y}_i = 10,302.9$.

```

> sreg_c <- summary(reg_c)
> yi_hat <- coef(reg_c)[[1]]+coef(reg_c)[[2]]*70
> yi_hat
[1] 10302.9

```

And standard error of $\hat{Y}_i = \sqrt{MSE \left(\frac{1}{n} + \frac{(X_i - \bar{X})^2}{S_{XX}} \right)} = 1001.231$

```

> alpha_d <- 0.1
> tc_d <- qt(1 - alpha_d/2, 50)
> sse_d <- sum(resid(reg_c)^2)
> #mean of price1 = 78.25
> #summary(data_tuna_b$price1)
> mse_d <- sse_d/50
> sxx_d <- sum((data_tuna_b$price1-78.25)^2)
> se_yi_hat_d <- sqrt(mse_d*(1/52+(70-78.25)^2/sxx_d))
> se_yi_hat_d
[1] 1001.231

```

- The critical value for a two-tailed 10% significance level with 50 degrees of freedom is $t_{0.05,50} = 1.6759$.

$$10,302.9 \pm (1.6759)1001.231 = [8,624.934, 11,980.87]$$

The 90% confidence interval is **[8,624.934, 11,980.87]**

```

> upper_bound <- yi_hat + tc_d*se_yi_hat_d
> lower_bound <- yi_hat - tc_d*se_yi_hat_d
> upper_bound
[1] 11980.87
> lower_bound
[1] 8624.934

```

- It means that we have 95% confident that the average number of cans sold

in a week will be between 8,624.934 and 11,980.87 when the price per can is 70 cents.

(e)

$$SAL1 = \beta_1 + \beta_2 PRICE1$$

$$\text{mean}(PRICE1) = 78.25$$

$$\text{mean}(SAL1) = 6718.712$$

$$\epsilon = \beta_2 \frac{PRICE1}{SAL1} = -5.0598$$

$$se(\epsilon) = se(\beta_2) \times \frac{PRICE1}{SAL1} = 0.9152$$

The critical value for a two-tailed 5% significance level with 50 degrees of freedom is $t_{0.025,50} = 2.0086$.

$$-5.0598 \pm (2.0086) 0.9152 = [-6.8981, -3.2215]$$

The 95% confidence interval is **$[-6.8981, -3.2215]$**

```
> cat("Elasticity:", ela, "\n")
Elasticity: -5.059825
> cat("95% CI: [", ci_lower, ", ", ci_upper, "]\n")
95% CI: [ -6.898149 , -3.221501 ]
```

The entire interval below -1. It means sales of brand no. 1 are fairly elastic .

=> sales are highly sensitive to price changes, indicating a high degree of elasticity.

There are many substitutes to canned tuna (other brands, other meats), so it is reasonable to expect demand to be elastic.

Consumers have plenty of substitutes to choose if the price of canned tuna increases. So the degree of elasticity is high.

(f)

① Null Hypothesis $H_0: elasticity = -3$

Alternative Hypothesis $H_1: elasticity \neq -3$

② significance level of 10%

③ We will test the hypothesis using a t-test.

$$\varphi = elasticity - (-3)/se(elasticity)$$

The test statistic is calculated as: $\varphi = (-5.059825 + 3)/0.9152 = -2.2506$

④ Rejection Region: $\{ t \leq -1.6759 \text{ or } t \geq 1.6759 \}$

the critical value for a two-tailed 10% significance level with 50 degrees of freedom is $t_{0.05,50} = 1.6759$.

⑤ $-2.25 \in RR$. Because t falls in rejection region, we **reject** the null hypothesis and conclude that the data do not support the claim.

There is an evidence that the elasticity is 3 is possibly not correct. And we conclude that the elasticity is possibly not -3.

p-value = 0.02884204

```
> print(paste("t-value =", round(t_statistic, 4)))
[1] "t-value = -2.2506"
> print(paste("t_critical=", round(t_critical, 4)))
[1] "t_critical= 1.6759"
> print(paste("p-value =", format(p_value, scientific = FALSE)))
[1] "p-value = 0.02884204"
```